So let's now begin our discussion on the actual topic of this article.

LXCOS

This is an operating system that is crafted using Chef and has the following features which help developer groups provision containers based on templates *(http://lxcos.io)*.

- It works in any environment and supports Amazon EC2 directly. You can run LXCOS as an Amazon instance, and then create containers on it and connect to them.
- It has pre-baked templates, mainly Web content management systems like Drupal, WordPress, LAMP, RoR, etc.
- You just have to specify the container name, template, memory and number of CPUs and you will have an isolated, accessible container started for you in less than a minute.

Getting started with LXCOS

LXCOS is released with an Apache licence and its code can be found at GitHub *(https://github.com/axelerant/lxcos)*. Let us go through the steps of having LXCOS installed in AWS and then create containers.

The requirements are:

- Chef server on AWS (installing Chef is out of scope for this article)
- Chef workstation (installing it is out of scope for this article)
- Knife ec2 wrapper on the Chef workstation

Remember, this might be the hard way, but there is a wrapper around the following steps to help you easily walk through the process, which I will discuss later.

Step 1: Clone the repository from GitHub mentioned above, into the Chef workstation:

```
git clone https://github.com/axelerant/lxcos.git
```

Step 2: Create a GitHub repo with your public ssh keys, which is safe. This provides you access to the newly created node on Amazon, or else we won't get access.

Create a file called *keys.git*, put all the keys that need access to the node in it, and push the file to a GitHub repository.

Step 3: In the code, modify Line 18 of */base/recipes/configure.rb* with the actual repo information:

```
repository "https://github.com/axelerant/admin-keys.git"
```

Step 4: Upload the cookbook to your Chef server:

```
knife cookbook upload -o /path/to/lxcos base
```

Step 5: Provision the node in AWS:

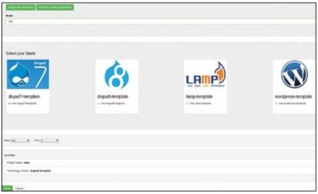

Figure 1: Create project

```
knife ec2 server create -r "recipe[base::install],recipe[base::
configure]" -I ami-7050ae18 --flavor t1.micro -G <security-group>
-Z <zone> -x ubuntu -S <keyname> -N <node-name> -i </path/to/
key> -K <<AMAZONSECRETKEY> -A <AMAZONACCESSKEY> --ebs-size G
```

Step 6: Make sure that you create a security group with 80 and 22 open and use that.

Step 7: Once the node is provisioned, you can log in to it as:

```
ssh goatos@<ec2-url>
```

Now you have a Chef powered node, which can spawn containers for you.

Here is an example:

```
goatos@lenient:~$ create_container.rb -n helloworld -t drupal7-
template -m 256M -c 0
{"name":"helloworld","ip_addr":["10.0.3.150"]}
goatos@lenient:~$
```

Building a PaaS solution and an example case study

All this can be automated with a gem called LXCOS Runner. You can find it at *https://github.com/axelerant/lxcos-runner*

If you use LXCOS Runner, you don't need to do any of the above stated steps manually, but you can still build a Web based PaaS. Let's go through a custom built internal PaaS application

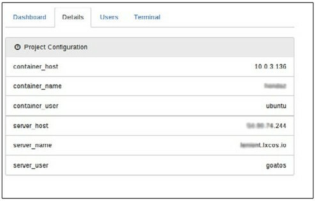

Figure 2: Container details